

Class 1

April 12, 2026

Course Details

Schedule

Class 1 Sunday	Class time: 3:00-6:00pm Microeconomics, Statistics, Operations	Monopoly game Trading game	Required game play: 7:00-9:00pm Assignment 2, due by 9am on April 13, 2026
Class 2 Monday	Class time: 3:00-6:00pm Operations, Financial Accounting	Production game	Required game play: 7:00-9:00pm Assignment 3, due by 9am on April 14, 2026
Class 3 Tuesday	Class time: 3:00-6:00pm Marketing, Finance, Strategy	Practice game	Required game play: 7:00-9:00pm Assignment 4, due by 9am on April 15, 2026
Class 4 Wednesday	Class time: 3:00-6:00pm Analyze Practice Game Prepare for Competition	Competition	Required game play: 7:00-9:00pm Assignment 5, due by 9am on April 16, 2026
Class 5 Thursday	Class time: 3:00-6:00pm Game Analysis	Competition	Required game play: 7:00-9:00pm No assignment tonight
Class 6 Friday	Class time: 11am-1pm Wrap Up	Wrap-Up Final Project	Assignment 6 and Final Project due by 11:59pm on May 3, 2026

Evaluation

Item	Individual or Group	Deadline (HK Time)	Points
Assignment 1	Individual	Sunday, April 12 by 3:00pm	10
Assignment 2	Group	Monday, April 13 by 9am	5
Assignment 3	Group	Tuesday, April 14 by 9am	10
Assignment 4	Group	Wednesday, April 15 by 9am	10
Assignment 5	Group	Thursday, April 16 by 9am	5
Assignment 6	Individual	Sunday, May 3 by 11:59pm	10
Final Project	Group	Sunday, May 3 by 11:59pm	30
Participation	Individual	Ongoing	15
Game Performance	Individual	Ongoing	5

Why ISM?

Why ISM?

- Student requests for a capstone course
- Application of MBA course concepts to practical situations
- Integration of MBA course topics
 - Complex business functions crossing boundaries of the disciplines by which we organize our curriculum

What is the Goal of the Firm?

What do firms do?

Strategy

Investing

Financing

Operating

“The Glue”

Leadership
Management
Negotiations
Decision Making

ISM: Some Strategic Decisions

- Where do we fit in this industry?
 - Low cost/low margin/high volume?
 - High cost/high margin/low volume?
 - First to market?
 - Fast follower?
 - What is our competitive advantage? Is it sustainable?
- What is our distribution strategy?
- What is our production strategy?
- What are our product, pricing and distribution strategies?
- What is our growth strategy?
- Should we partner with other teams?
- How should we structure our management team for optimal performance?

ISM: Some Investing Decisions

- Should we buy a new factory or DC?
- Should we add capacity to a current factory?
- Should we add daily recurring expenses to our factory?
- Should we build a new product?
- Should we shut down a factory or DC?

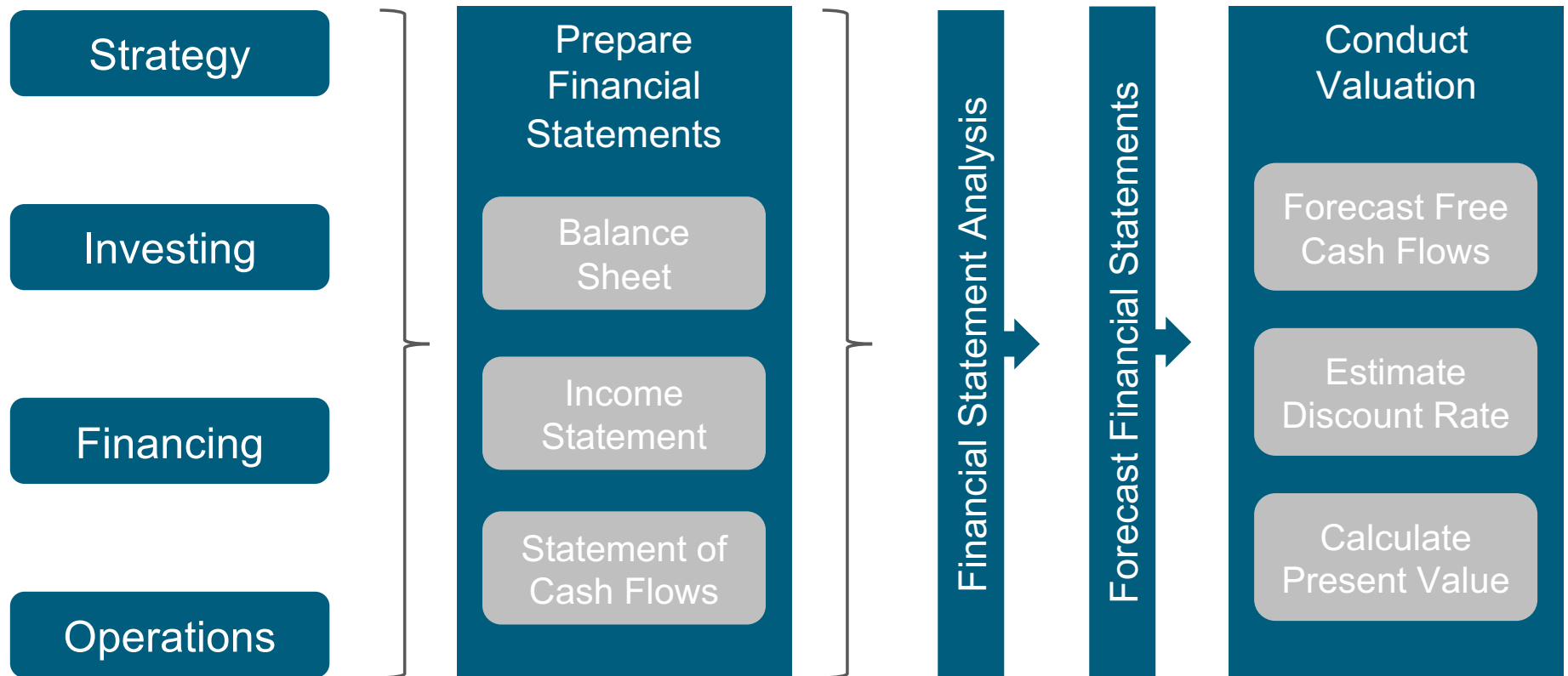
ISM: Some Financing Decisions

- Should we issue bonds?
- How can we manage our working capital to help us finance the running of the business?
 - Supplier terms
- Should we retire debt?
- Should we pay dividends?

ISM: Some Operating Decisions

- Which products to produce?
 - R&D, focus groups, product architecture, roll-out
- How do we **avoid/manage stock-outs**?
- How do we price our products over time?
 - Retail
 - Wholesale
- How do we **manage idle time**?
- How do we ship our goods?
- What is our optimal reorder point and order quantity?
- How do we manage our partner relationships? Competitors?

Big Picture: Valuation



Notes

Micro, Statistics, Operations

Pricing, Demand, and Reorder Points

Monopoly Game

- You will need the monopoly game case brief posted on Canvas.
- What is the optimal price to charge for your product?
- What is expected demand at that price?
- What is the optimal reorder point?

Assumptions

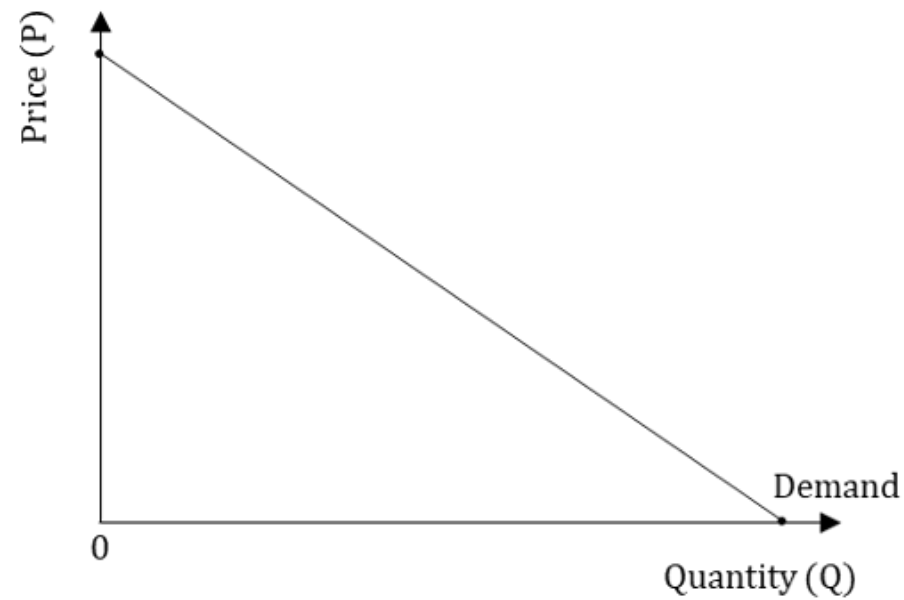
Maximum WTP	\$500
Minimum WTP	\$0
Materials Cost	\$100
Market Size	300,000 potential buyers
Arrival Rate	0.0001
Lead Time	3.5 days
Opportunity cost of capital	10%

WTP = Willingness to pay for the product

WTP is uniformly distributed on the range of minimum to maximum WTP

There is no variation in lead time.

Demand Curve



Micro: Profit Maximization and Optimal Price

Monopolist

Maximum WTP	\$500
Minimum WTP	\$0
Materials Cost	\$100
Market Size	300,000 potential buyers, footfall = $300,000 * .0001 = 30/\text{day}$
Optimal Price	\$_____

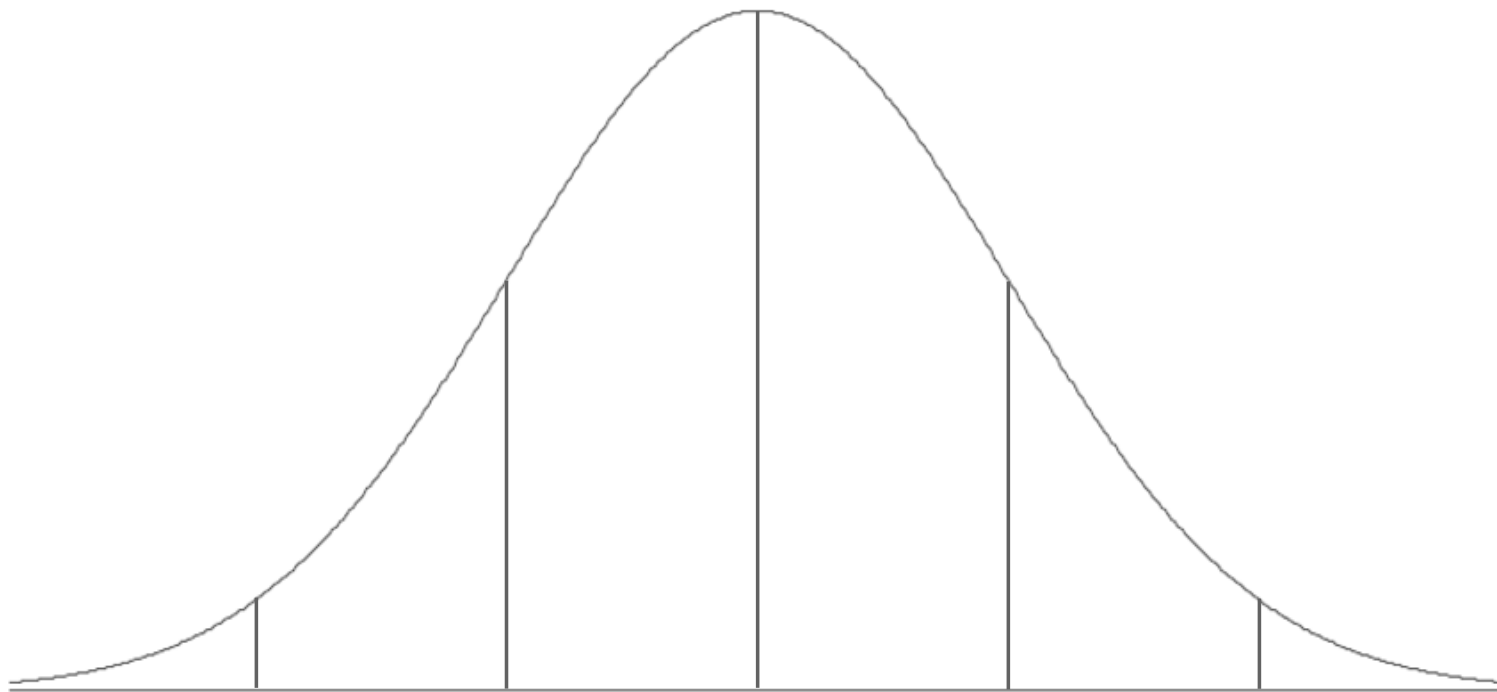
Statistics: Expected Daily Sales

- Expected value of a binomial: $E(x) = np$
- Variance of a binomial: $\text{Var}(x) = np(1 - p)$
- Standard deviation of a binomial: $S(x) = \sqrt{np(1 - p)}$

Expected daily sales and standard deviation for our product:

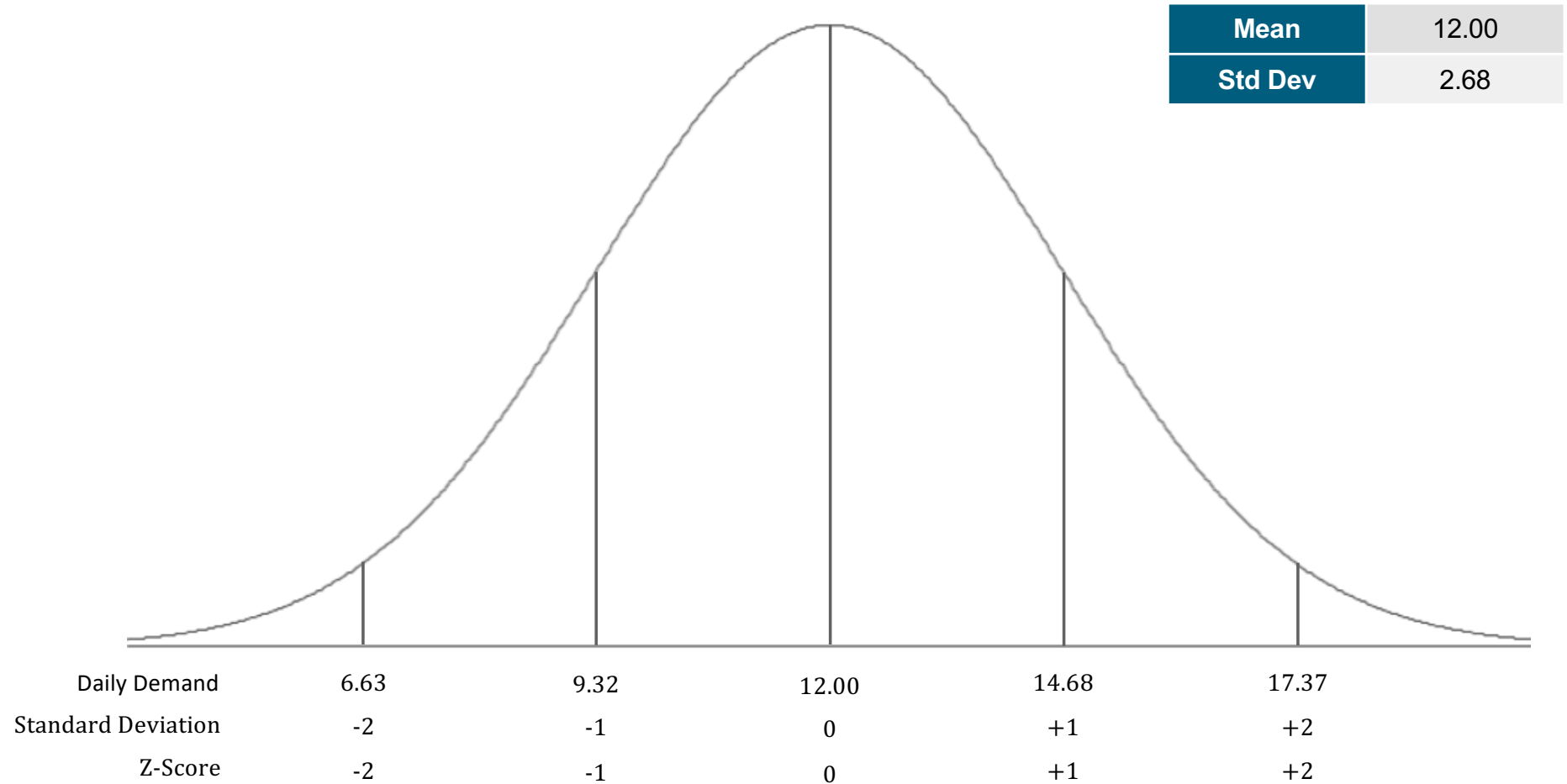
Statistics: Expected Sales (Demand)

Distribution



Optimal Reorder Point Example

Daily demand



How do we determine our Reorder Point?

What is a reorder point?

- The reorder point is the level of inventory at which we order more units.

How do we determine our Reorder Point?

Lead time demand and safety stock

Reorder Point = Lead time demand + Safety stock

$$\text{Reorder Point} = (w * \mu_d) + (z * \sigma_d * \sqrt{w})$$

w: lead time

μ_d : average number of units demanded in each period

z: z-score associated with the critical fractile

σ_d : standard deviation of demand in each period

Lead time demand is the number of units that we expect to sell after we place an order for more inventory but before it arrives.

$$\text{Lead Time Demand} = w * \mu_d$$

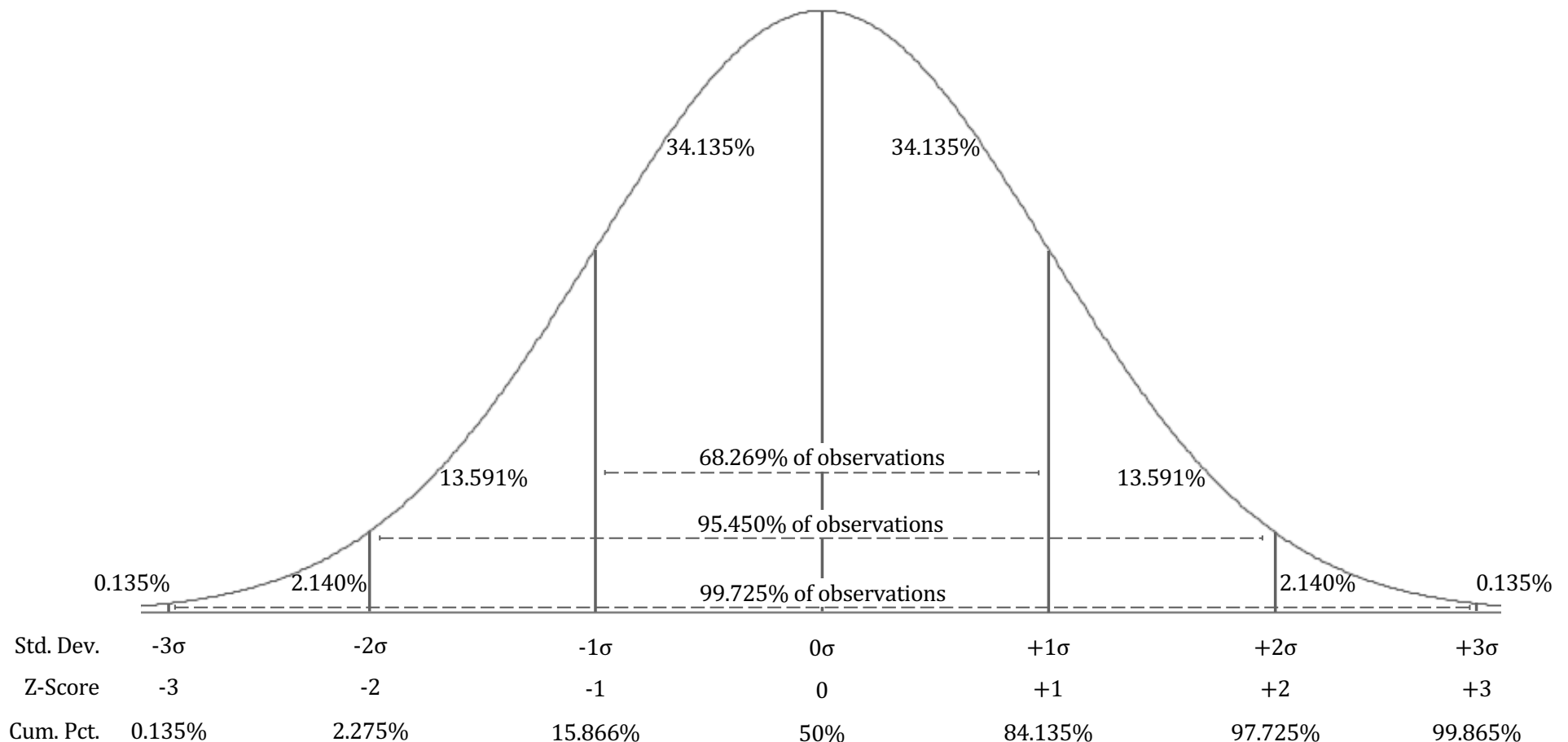
Safety stock is the additional number of units we should hold in inventory to satisfy demand up to the critical fractile, taking into account the variability of demand.

$$\text{Safety Stock} = z * \sigma_d * \sqrt{w}$$

What % of the time do we want to satisfy demand?

Probability functions

To determine the reorder point, we need to know what % of the time we want to be able to satisfy demand.



What % of the time do we want to satisfy demand?

Cost of under and overstocking

The cost of **understocking (C_u)** is the cost of being a unit short

- Lost sales
- **Lost contribution margin**
- Too little of the right stuff
- The expected gain from ordering one more unit from Q to $Q+1$ is the cost of understocking multiplied by the probability of demand being greater than Q

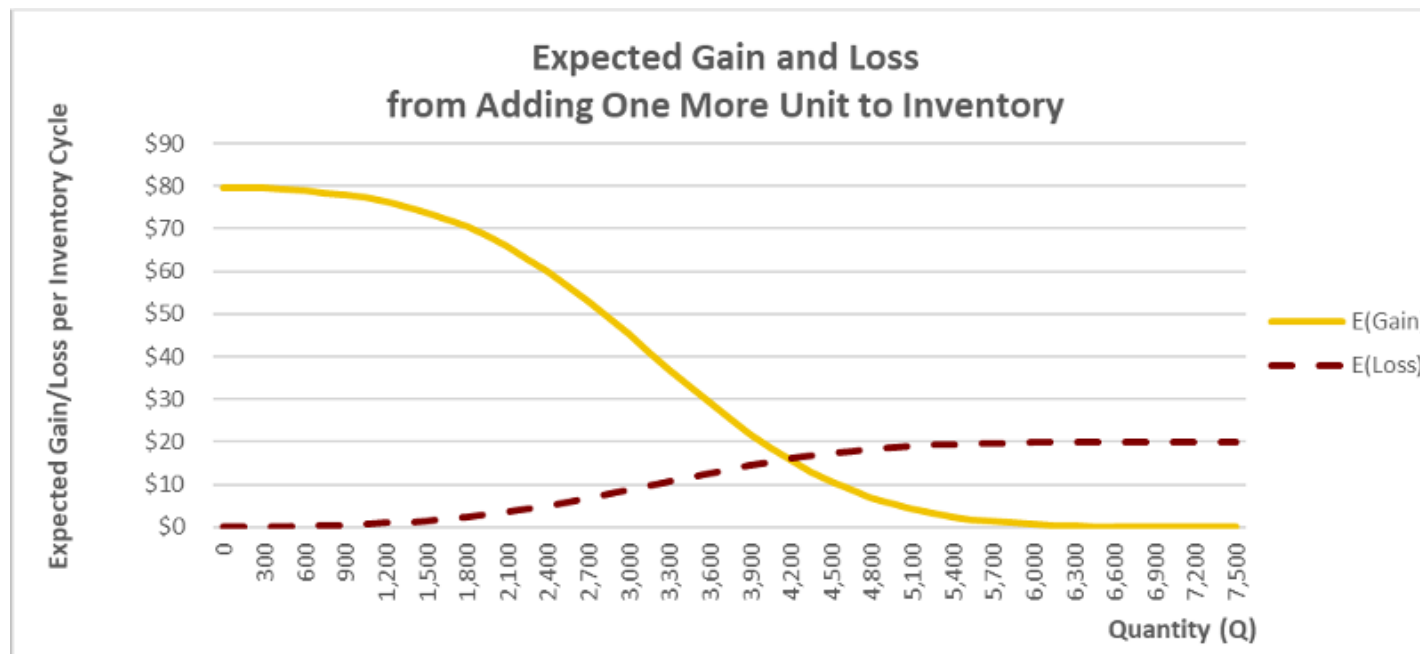
The cost of **overstocking (C_o)** is the opportunity cost of holding an extra unit

- Leftover inventory
- Too much of the wrong stuff
- The expected loss from ordering one more unit from Q to $Q+1$ is the cost of overstocking multiplied by the probability of demand being less than or equal to Q

What % of the time do we want to satisfy demand?

Choose Q to maximize profits

Need to trade off the expected gain from ordering one more unit with the expected loss from ordering one more unit.



What % of the time do we want to satisfy demand?

Should we add one more unit to our reorder point?

Expected Gain from ordering one more unit

$$C_u * P(\text{Demand} > Q)$$

$$C_u * (1 - P(\text{Demand} \leq Q))$$

$$C_u * (1 - F(Q))$$

Expected Loss from ordering one more unit

$$C_o * P(\text{Demand} \leq Q)$$

$$C_o * F(Q)$$

Set equal to each other

$$C_u * (1 - F(Q)) = C_o * F(Q)$$

Solve for $F(Q)$

$$\frac{C_u}{(C_u + C_o)}$$

Solve for Q

$$F^{-1}\left(\frac{C_u}{(C_u + C_o)}\right)$$

Find Q such that $P(\text{Demand} \leq Q) = \text{critical fractile}$

How much demand do we want to satisfy?

Finding Q from the critical fractile

$$\text{Critical fractile} = \frac{C_u}{(C_u + C_o)}$$

Use a standard normal probability table to find the z-score that corresponds to the probability, or use the EXCEL function:

=NORM.INV(probability, mean, standard deviation)

where probability = critical fractile, mean = 0, and standard deviation = 1.

z	0.00	0.01	0.02
-0.5	0.3085	0.3050	0.3015
-0.4	0.3446	0.3409	0.3372
-0.3	0.3821	0.3783	0.3745
-0.2	0.4207	0.4168	0.4129
-0.1	0.4602	0.4562	0.4522
-0.0	0.5000	0.4960	0.4920

z	0.00	0.01	0.02
0.0	0.5000	0.5040	0.5080
0.1	0.5398	0.5438	0.5478
0.2	0.5793	0.5832	0.5871
0.3	0.6179	0.6217	0.6255
0.4	0.6554	0.6591	0.6628
0.5	0.6915	0.6950	0.6985

Reorder Point

Example, where our optimal price = \$300, MC = \$100, $r = 10\%$

- What is the cost of understocking?
- What is the cost of overstocking?
- What is the critical fractile?
- What is the z-score?
- What is our lead time demand?
- What is our safety stock?
- What is our reorder point?

Cost of Understocking

C_u = cost of under stocking

- Cost of being a unit short
- If the reorder point is one unit too low, we lose a sale every sales cycle

What is the cost of understocking our product?

Cost of Overstocking

C_o = cost of over stocking

- Opportunity cost of holding an extra unit
- If the reorder point is one unit too high, always holding one unit too many

What is the cost of overstocking our product?

Critical Fractile

What is the critical fractile for this product?

$$\text{Critical fractile} = \frac{C_u}{(C_u + C_o)}$$

Z-Score

What is the z-score that corresponds to the critical fractile?

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

=NORM.INV(probability, mean, standard deviation) = NORM.INV(0.9988566, 0, 1) = 3.0502

Reorder Point

What is our optimal reorder point with lead time demand?

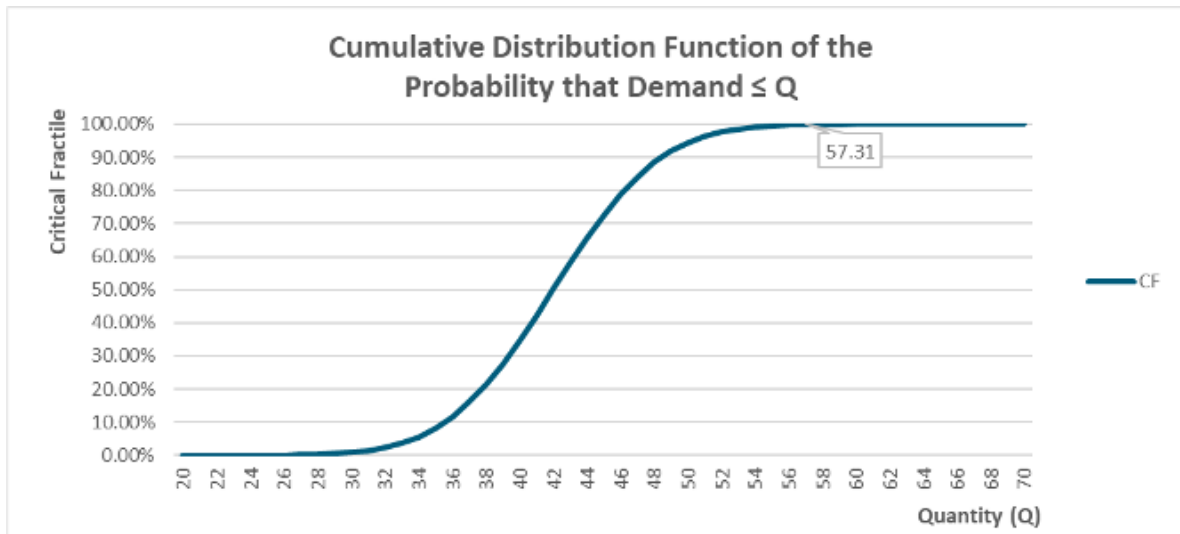
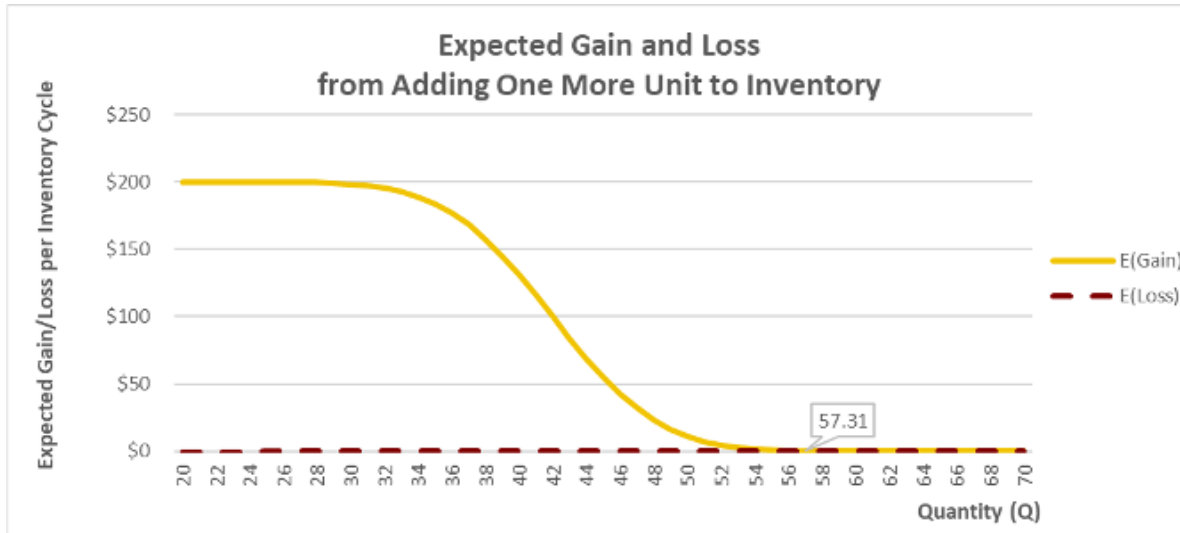
Lead Time Demand:

Safety Stock:

Reorder Point:

Optimal Reorder Point

Finding Q from the critical fractile – with lead time

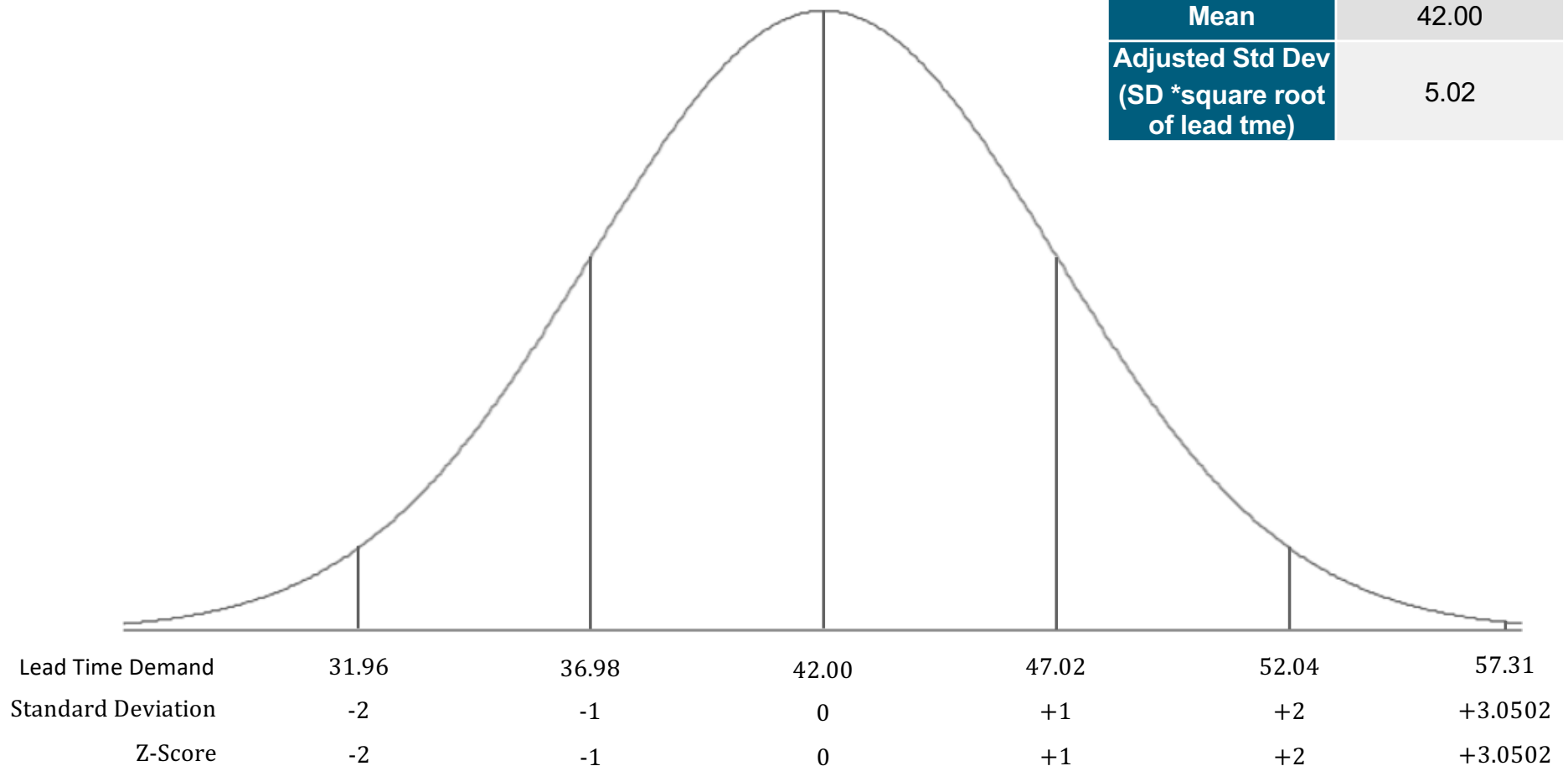


Q	CF	Z-score	E(Gain)	E(Loss)	E(Profit)
40	34.516416%	-0.3984	\$130.97	\$0.08	\$130.89
41	42.105128%	-0.1992	\$115.79	\$0.10	\$115.69
42	50.000000%	0.0000	\$100.00	\$0.11	\$99.89
43	57.894872%	0.1992	\$84.21	\$0.13	\$84.08
44	65.483584%	0.3984	\$69.03	\$0.15	\$68.88
45	72.495134%	0.5976	\$55.01	\$0.17	\$54.84
46	78.722194%	0.7968	\$42.56	\$0.18	\$42.38
47	84.038072%	0.9960	\$31.92	\$0.19	\$31.73
48	88.400114%	1.1952	\$23.20	\$0.20	\$23.00
49	91.840661%	1.3944	\$16.32	\$0.21	\$16.11
50	94.449145%	1.5936	\$11.10	\$0.22	\$10.89
51	96.350098%	1.7928	\$7.30	\$0.22	\$7.08
52	97.681709%	1.9920	\$4.64	\$0.22	\$4.41
53	98.578324%	2.1913	\$2.84	\$0.23	\$2.62
54	99.158630%	2.3905	\$1.68	\$0.23	\$1.46
55	99.519649%	2.5897	\$0.96	\$0.23	\$0.73
56	99.735536%	2.7889	\$0.53	\$0.23	\$0.30
57.31	99.885662%	3.0502	\$0.23	\$0.23	\$0.00
58	99.928190%	3.1873	\$0.14	\$0.23	(\$0.09)
59	99.964602%	3.3865	\$0.07	\$0.23	(\$0.16)
60	99.983190%	3.5857	\$0.03	\$0.23	(\$0.20)
61	99.992311%	3.7849	\$0.02	\$0.23	(\$0.21)
62	99.996613%	3.9841	\$0.01	\$0.23	(\$0.22)
63	99.998563%	4.1833	\$0.00	\$0.23	(\$0.23)
64	99.999413%	4.3825	\$0.00	\$0.23	(\$0.23)
65	99.999769%	4.5817	\$0.00	\$0.23	(\$0.23)
66	99.999913%	4.7809	\$0.00	\$0.23	(\$0.23)
67	99.999968%	4.9801	\$0.00	\$0.23	(\$0.23)
68	99.999989%	5.1793	\$0.00	\$0.23	(\$0.23)
69	99.999996%	5.3785	\$0.00	\$0.23	(\$0.23)
70	99.999999%	5.5777	\$0.00	\$0.23	(\$0.23)

Optimal Reorder Point

With lead time

Mean	42.00
Adjusted Std Dev (SD *square root of lead tme)	5.02



Relevant Formulas for Monopoly Game

Demand Function	$Q^d = a - bP$
Demand Curve	$P = \frac{a}{b} - \frac{1}{b}Q^d$
Max WTP	$\text{Max WTP} = \frac{a}{b}$
Optimal Price	$P = \frac{\text{Max WTP} + \text{Marginal Cost}}{2}$
Expected Daily Demand	$E(Q_{\text{daily}}^d) = np = \text{Daily Arrivals} * \text{Probability of Buying}$
Standard Deviation of Daily Demand	$S(Q_{\text{daily}}^d) = \sqrt{np(1-p)}$
Cost of Overstocking	$C_o = \text{Materials Cost} * \text{Annual Interest Rate}$
Cost of Understocking	$C_u = \text{Contribution Margin per Unit} * \# \text{ of Inventory Cycles per Year}$ $C_u = (\text{Price} - \text{Materials Cost}) * \frac{\text{Expected Daily Sales} * \text{Days per Year}}{\# \text{ of Units per Inventory Cycle (Batch Size)}}$
Critical Fractile	$\text{Critical Fractile} = \frac{C_u}{(C_u + C_o)}$
Optimal Reorder Point	$\text{Reorder Point} = (w * \mu_d) + (z * \sigma_d * \sqrt{w}) = \text{Lead Time Demand} + \text{Safety Stock},$ where w is lead time, μ_d is average daily demand, z is z-score, σ_d is standard deviation of demand

Notes

Notes

Monopoly Game

Teams

1	B612	1-25	Masa Ishigaki
1	B612	1-25	Takeshi Tanaka
1	B612	1-25	Hyeyoung Lee
1	B612	1-25	Valerii Egorov
1	B612	1-25	Carlos Naibryf
	B612	1-25	Fengshu Jin
2	Dune	1-26	Jo Hayes
2	Dune	1-26	Jenny Yang
2	Dune	1-26	Morry Mori
2	Dune	1-26	Suliya Suliya
2	Dune	1-26	Betty Wang
2	Dune	1-26	Kosuke Okura
3	Globex	1-27	Prashanth Palepu
3	Globex	1-27	Lisa Lau
3	Globex	1-27	George Chia
3	Globex	1-27	Jeffrey Chen
3	Globex	1-27	Kacey Du
3	Globex	1-27	Lambert Xu
4	Gotham	1-28	Inge Supatra
4	Gotham	1-28	Bryan Wong
4	Gotham	1-28	Delphine Terrien
4	Gotham	1-28	Benjamin Jiang
4	Gotham	1-28	Dai Kato
4	Gotham	1-28	Ngiap Seng Khoo

5	Panem	1-29	Chris Ma
5	Panem	1-29	Shiyuan Tian
5	Panem	1-29	Yohei Nakadate
5	Panem	1-29	Derrick Teo
5	Panem	1-29	Jack Meng
5	Panem	1-29	Jason Weng
6	Vulcan	1-31	Eric Zhang
6	Vulcan	1-31	Jony Hu
6	Vulcan	1-31	Laurence Zhu
6	Vulcan	1-31	Yehwan Kim
6	Vulcan	1-31	Tom Hsieh
6	Vulcan	1-31	Sudeep Rathee
7	Westeros	1-32	Kosuke Shinohara
7	Westeros	1-32	Ken Ng
7	Westeros	1-32	Taku Yasuda
7	Westeros	1-32	Ryan Kim
7	Westeros	1-32	Andy Yoo
7	Westeros	1-32	Jumpei Maruyama
8	Zion	1-33	Ken Chew
8	Zion	1-33	Ryo Aikawa
8	Zion	1-33	Louis Woenardi
8	Zion	1-33	Dimas Purnama
8	Zion	1-33	Chris Kwan
8	Zion	1-33	Alex Wang

Block A 101 Map



Monopoly Game

- Highlights

- You have 2 products – Hormone and a specialty good
- All products cost \$100 for materials + ~\$0.46 in allocated factory depreciation per unit to manufacture
- Factories run 24 hours a day
- All products are made in batches of 100 units
- It takes 2 ½ days to produce a batch
- It takes 24 hours to ship a batch from the factory to your DC
- Shipping prices: Free within region
- Market size is 300,000 for the hormone product and 140,000 for the specialty product
- All sales are made from your DC
- Retail customers arrive your DC at the rate of $0.0001 \times \text{the market size per day}$
- All inventory becomes obsolete at the end of the game

- Goals

- Maximize Cash
- Optimize retail prices and reorder points
- Learn Interface: Reports – financials and others
- Learn Interface: Chat

Monopoly Game Map

The link to the game is on the Canvas home page in the Day 1 section

Gleacher

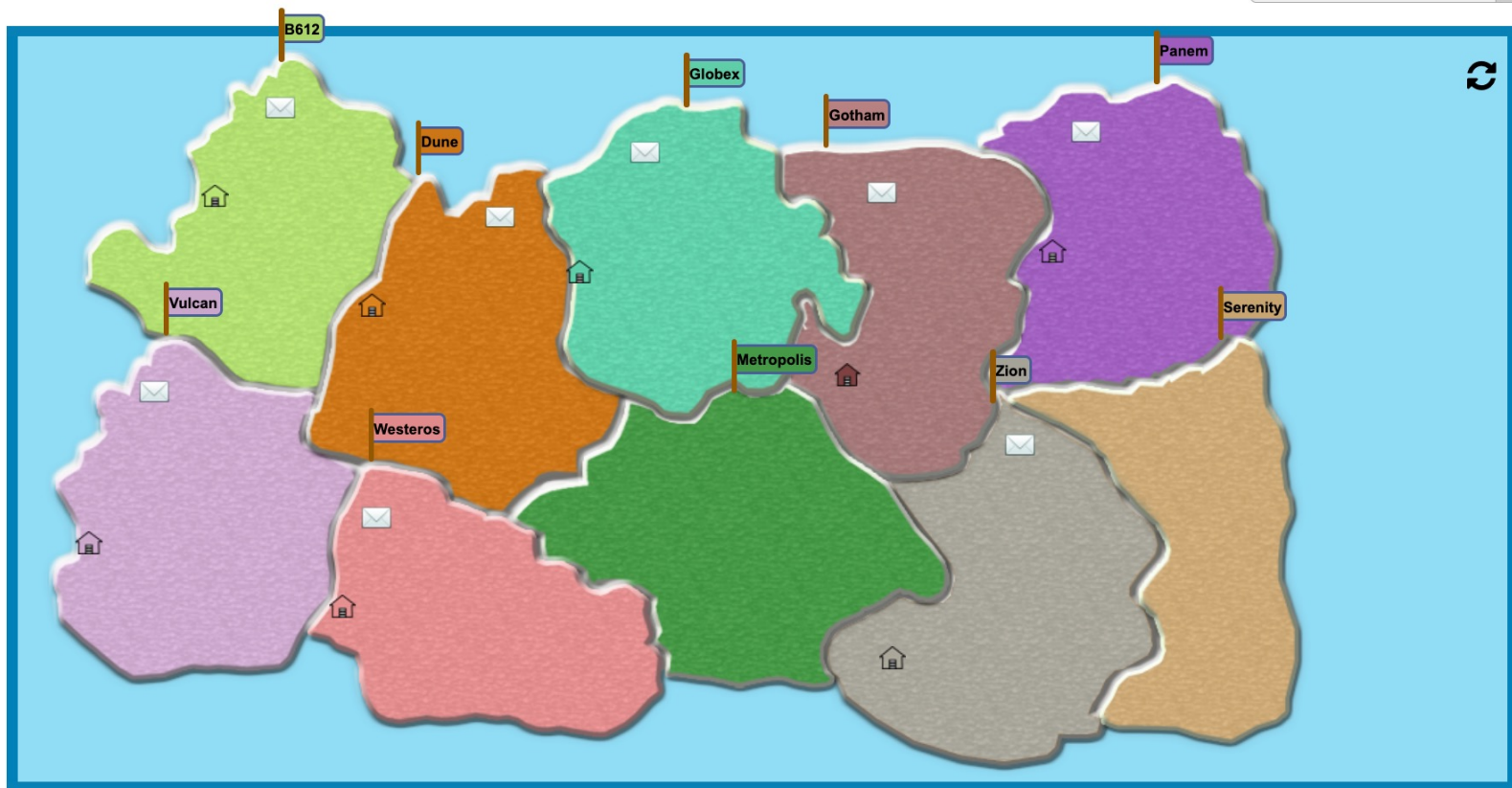
ADMIN VIEW

DAY: 91

STATUS: SUSPENDED



Open Report



Scoreboard

Rank	World	Team Name	◆	Cash Position
1		B612		<u>\$ 453,336</u>
1		Dune		<u>\$ 453,336</u>
1		Globex		<u>\$ 453,336</u>
1		Gotham		<u>\$ 453,336</u>
1		Panem		<u>\$ 453,336</u>
1		Vulcan		<u>\$ 453,336</u>
1		Westeros		<u>\$ 453,336</u>
1		Zion		<u>\$ 453,336</u>

Monopoly Game

- Maximum willingness to pay for each product
- Optimal retail price for each product
- Expected daily demand for each product
- Expected daily standard deviation of demand for each product
- Cost of over- and under-stocking each product
- Critical fractile for each product
- Lead time demand for each product
- Safety stock for each product
- Optimal reorder point for each product

Monopoly Game

- Start at the end of quarter 1, begin on day 92
- 60 minutes to play to the end of quarter 4, end on day 364
- Speed: 273 days over 60 minutes is approximately 4.55 game days per minute.

As of day 91.61

Name : Local DC in Microsoft
Owner : Microsoft
Location : Microsoft
Status : READY
Operating income : \$ 332,733

Total inventory : 352 units
On-hand inventory : 252 units
Price range : \$ 400 to \$ 400
Total retail sales : 1,242 units

2 Inventories

10 records

Include Inactive: ☐ Search:

Product	\$	Units	Last date	
Name	Price	On hand	Any	
Circulation	400	162	90.99	
Hormone	400	90	90.77	

Showing 1 to 2 of 2 entries

1

2 Incoming Shipping Agreements

10 records

Search:

Id	Origin	Product	Reorder point	Terms	Days delayed	Batches	Projected contribution per unit Margin	Status	
		Name							
1		Hormone	100	Container 100@\$ 100.0	0	7	\$ 300	ACTIVE	
2		Circulation	100	Container 100@\$ 100.0	0	9	\$ 300	ACTIVE	

Showing 1 to 2 of 2 entries

1

0 Outgoing Shipping Agreement

ADD NEW +

Retail Price Response for Macrosoft-Hormone (H+) in Local DC in Macrosoft

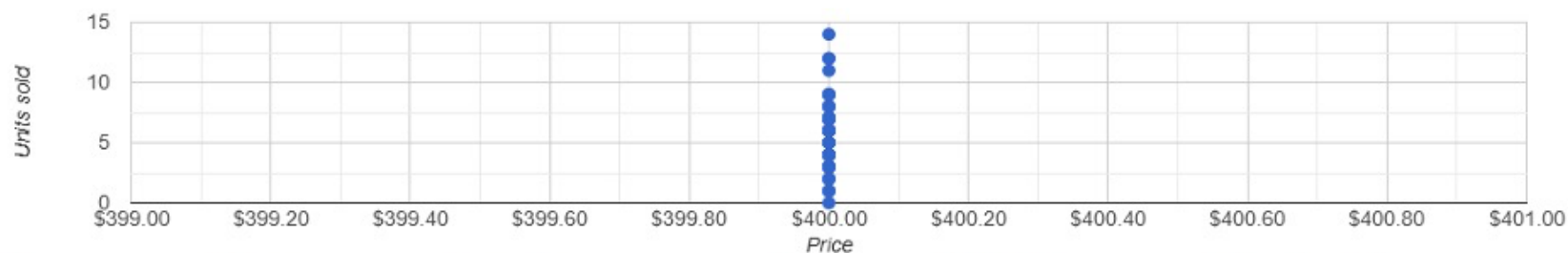
Note: Intervals when there is no on-hand inventory are ignored.

Currently 93 units of on-hand inventory priced at \$400 

* All rates are unconsolidated

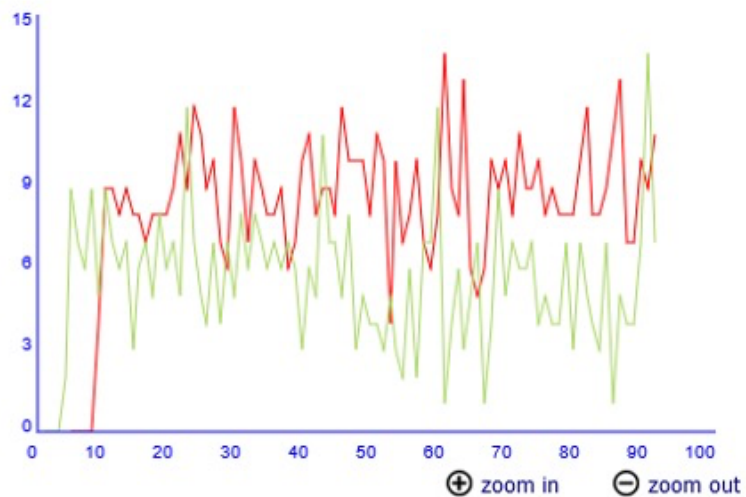
Price 400.00  400.00 Interval days 00.00  01.00 Start day 03.50  91.00

Domain : Plot : Trendline :



Start day	End day	Interval days	Price	Units sold	Units sold per day	
91	91	0	\$400	0	0	0
90	91	1	\$400	7	7	7
89	90	1	\$400	14	14	14
88	89	1	\$400	7	7	7
87	88	1	\$400	4	4	4
86	87	1	\$400	4	4	4
85	86	1	\$400	5	5	5
84	85	1	\$400	1	1	1
83	84	1	\$400	7	7	7
82	83	1	\$400	3	3	3

Daily Retail Sales from Local DC in Microsoft



drag mouse over region of interest to zoom in plot

Reset Data Clear

■ Microsoft-Circulation ■ Microsoft-Hormone

Daily Retail Sales from Local DC in Microsoft



Time period in Days	Microsoft- Circulation	Microsoft- Hormone
1		0
2		0
3		0
4		2
5	0	9
6	0	7
7	0	6
8	0	9
9	4	5
10	9	9
11	9	7
12	8	6

Reset Chart Clear

■ Microsoft-Circulation ■ Microsoft-Hormone

Financial Statement for : Microsoft

Income Statement Quarter 1 to 1

Consolidated Statements of Operations

Click on \$ to view values as a percent of revenues net adjustments. Click on an account name to open a chart. Click on a value to view details.

	\$	Quarter 1
Sales revenue	\$	496,800
Cost of goods sold	\$	124,769
Gross profit	\$	372,031
Total selling expense	\$	0
Distribution center depreciation expense	\$	41,667
Idle factory expense	\$	960
Other operating expenses	\$	42,627
Operating income	\$	329,404
Interest revenue	\$	2,811
Net interest and fees revenue (expense)	\$	2,811
Total net income before tax	\$	332,215
Income tax expense	\$	116,275
Net income after tax and before extraordinary items	\$	215,940
Net income	\$	215,940



Financial Statement for : Microsoft

Income Statement Quarter 1 to 1

Consolidated Statements of Operations

Click on \$ to view values as a percent of revenues net adjustments. Click on an account name to open a chart. Click on a value to view details.

	%	Quarter 1
Sales revenue	%	100
Cost of goods sold	%	25.11
Gross profit	%	74.89
Total selling expense	%	0
Distribution center depreciation expense	%	8.39
Idle factory expense	%	0.19
Other operating expenses	%	8.58
Operating income	%	66.31
Interest revenue	%	0.57
Net interest and fees revenue (expense)	%	0.57
Total net income before tax	%	66.87
Income tax expense	%	23.40
Net income after tax and before extraordinary items	%	43.47
Net income	%	43.47



Financial Statement for : MicrosoftBalance Sheet Quarter 1 to 0

Consolidated Balance Sheets

Click on \$ to view values as a percent of total assets. Click on an account name to open a chart. Click on a value to view details.

Assets	\$	Quarter 1	Quarter 0
Current assets:			
Cash	\$	453,336	200,000
Work in process inventory	\$	10,020	10,000
Finished goods inventory	\$	25,918	0
Tax benefit	\$	0	0
Total current assets	\$	489,273	210,000
Noncurrent assets:			
Land	\$	200,000	200,000
Property plant and equipment - factories:			
Property plant and equipment - factories	\$	100,000	100,000
Less accumulated depreciation - factories	\$	(1,667)	0
Net Property plant and equipment - factories	\$	98,333	100,000
Property plant and equipment - distribution centers:			
Property plant and equipment - distribution centers	\$	2,500,000	2,500,000
Less accumulated depreciation - distribution centers	\$	(41,667)	0
Net Property plant and equipment - distribution centers	\$	2,458,333	2,500,000
Total assets	\$	3,245,940	3,010,000
Liabilities and shareholders' equity			
Liabilities:			
Current liabilities:			
Raw materials payable	\$	30,000	10,000
Taxes payable	\$	0	0
Other payables	\$	0	0
Total current liabilities	\$	30,000	10,000
Noncurrent liabilities:			
Noncurrent liabilities	\$	0	0
Shareholder equity:			
Contributed capital	\$	3,000,000	3,000,000
Retained earnings	\$	215,940	0
Total shareholder equity	\$	3,215,940	3,000,000
Total liabilities and shareholders' equity	\$	3,245,940	3,010,000

Financial Statement for : MicrosoftBalance Sheet Quarter 1 to 0

Consolidated Balance Sheets

Click on \$ to view values as a percent of total assets. Click on an account name to open a chart. Click on a value to view details.

Assets	%	Quarter 1	Quarter 0
Current assets:			
Cash	%	13.97	6.64
Work in process inventory	%	0.31	0.33
Finished goods inventory	%	0.80	0
Tax benefit	%	0	0
Total current assets	%	15.07	6.98
Noncurrent assets:			
Land	%	6.16	6.64
Property plant and equipment - factories:			
Property plant and equipment - factories	%	3.08	3.32
Less accumulated depreciation - factories	%	-0.05	0
Net Property plant and equipment - factories	%	3.03	3.32
Property plant and equipment - distribution centers:			
Property plant and equipment - distribution centers	%	77.02	83.06
Less accumulated depreciation - distribution centers	%	-1.28	0
Net Property plant and equipment - distribution centers	%	75.74	83.06
Total assets	%	100	100
Liabilities and shareholders' equity			
Liabilities:			
Current liabilities:			
Raw materials payable	%	0.92	0.33
Taxes payable	%	0	0
Other payables	%	0	0
Total current liabilities	%	0.92	0.33
Noncurrent liabilities:			
Noncurrent liabilities	%	0	0
Shareholder equity:			
Contributed capital	%	92.42	99.67
Retained earnings	%	6.65	0
Total shareholder equity	%	99.08	99.67
Total liabilities and shareholders' equity	%	100	100

Monopoly Game Solutions

Price, Lead Time Demand, Safety Stock, Reorder Point

Hormone

Optimal Price =

Lead Time Demand =

Safety Stock =

Reorder Point =

Specialty

Optimal Price =

Lead Time Demand =

Safety Stock =

Reorder Point =

Monopoly Results & Trading Game

Tonight – see you in the classroom at 7pm

Monopoly Game Results



Winner!

Monopoly Game Solutions

Price, Lead Time Demand, Safety Stock, Reorder Point

Hormone

Optimal Price =

Lead Time Demand =

Safety Stock =

Reorder Point =

Specialty

Optimal Price =

Lead Time Demand =

Safety Stock =

Reorder Point =

Trading Game

- Highlights

- You have 2 products – Hormone and a specialty good
- All products cost \$100 for materials + ~\$0.46 in allocated factory depreciation per unit to manufacture
- Factories run 24 hours a day
- All products are made in batches of 100 units
- It takes 2 ½ days to produce a batch
- Shipping within region: it takes 24 hours to ship a batch from the factory to your DC, and shipping is free
- Shipping between regions: 3 days and \$40/unit by mail; 21 days and \$10,000 per container
 - Mail is shipped in sets of 10 units and each container holds up to 1,000 units
- Market size for the hormone product is 300,000 in each region
- Market size for the specialty product is 140,000 in its home region and 60,000–200,000 in other regions
- All sales are made from your DC
- Retail customers arrive your DC at the rate of $0.0001 \times \text{the market size per day}$
- All inventory becomes obsolete at the end of the game

- Goals

- Maximize Cash
- Practice trading with other teams
- Optimize wholesale prices and supply chains
- Learn Interface: Reports – financials and others
- Learn Interface: Chat

Things we will encounter

- Pricing
 - Pricing for retail and wholesale customers
 - Double marginalization problem
- Supply Chain Risks
 - Demand side
 - Supply side
- Capacity
 - Excess capacity
- Shipping Agreement Terms
- Shipping
 - Mail vs. container
- Stockouts
- Reorder Points
- Reports

Trading Game

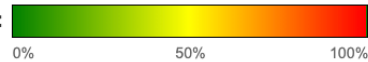
Class Play

- 60 minutes of play – we begin on day 92 and end on day 364
 - Speed: $273/60 = 4.55$ simulated days per minute
- Assigned study rooms
 - Take time to come up with a plan
 - Watch your inventory and your reports
- Negotiate directly, through chat or walking to rooms, or by proposing shipping agreements
- Pay particular attention to:
 - Reorder points for both retail and wholesale goods
 - Shipping methods and costs
 - Pricing
 - Demand and capacity issues
 - Margins
 - Stock outs
 - Idle time (should not be any!)
 - Batch size
 - Financial statements and working capital
 - Inventory status report
 - Product analysis report
 - Market grid
 - Shipping agreement terms
- Kathleen will play Serenity and Catherine will play in Metropolis. We are happy to trade!

Market Grid

	B612	Dune	Globex	Gotham	Metropolis	Panem	Serenity	Vulcan	Westeros	Zion	Total
Hormone	507	507	507	507	507	507	507	507	507	507	5,070
Circulation	735										735
Cardiac		735									735
Cancer			735								735
Gas				735							735
Thyroid					735						735
Opiate								735			735
Motion						735					735
Proteins							735				735
Botulinum									735		735
Amphetamine										735	735
Total	1,242	1,242	1,242	1,242	1,242	1,242	1,242	1,242	1,242	1,242	12,420

Fraction of total customers served :



Trade with others to use your excess capacity

Double Marginalization Problem

How to Trade!

New Shipping Agreement

Destination : Local DC in Hamazon [Hamazon]

Product : Macrosoft-Circulation

Revision : 1

Created by : admin on day 92

Orders

Order quantity : 100

Unit price (\$) : 400.0

Days sales outstanding : 0

Ship mode : Container

Ship time : 21 days

Cost to ship 1,000 units : \$ 10,000

Cost to ship 100 units : \$ 10,000

Franchise fees and royalties

Fees paid by Local DC in Hamazon to Local DC in Macrosoft

One time at inception (\$) : 0.00

Daily until termination (\$) : 0.00

Upon cash receipt from each retail customer : \$ 0.00 per unit plus 0.00 % of retail price.

Slotting fees and buydowns

Fees paid by Local DC in Macrosoft to Local DC in Hamazon

One time at inception (\$) : 0.00

Daily until termination (\$) : 0.00

Limits

☐ Priority :

☐ Reorder point :

☐ Retail price :

Termination and returns

May terminate at will after 0 days

Early termination fee (\$) : 0

☐ Returns allowed ?

Penalty for late shipments

0.00 % per day of order price,

starting 0 days after shipping agreement is triggered, until either a batch is shipped or discount reaches full order price.

Other terms

Other terms

Comment

Comment

Save

SAVE AND PROPOSE

Cancel

Reports

Start learning what is in each report – these are the reports for today

Open Report

Finance and Accounting

Financial Statements

Working capital analysis

Marketing

Market grid

Price response

Supply Chain

Inventory views

Inventory status

Inventory grid

Inventory status grid

Shipping Agreements - all

Active shipping agreement sankey

Good Luck!!

Price!

Idle
Time!

Chat!

Reports!

Accept
Deals

Mail!

Trade!

Reorder
Points!

Stock
outs!

Next Steps

- Prepare Assignment 2
 - This assignment is short and most of it you have already completed in the class.
 - Please spend time exploring the production game interface and reading/reviewing the documents below
- Read Gleacher Game Case Brief
- Review Product Game Market Research Report
- Review Player Manual

Thanks!